Session 6: Memory and problem solving

BSCAI 2026
Javier Muro

- How do humans learn from limited examples? Theories of inductive reasoning and attention mechanisms.
- The challenge of generalization in AI: Overfitting, data augmentation
- Why deep networks often require massive amounts of data to generalize?

Types of Learning

- Reinforcement learning
  - Learning through rewards in uncertain environments

- Unsupervised learning
  - Learn statistical regularities and build internal categories automatically i.e. faces, voices stand out

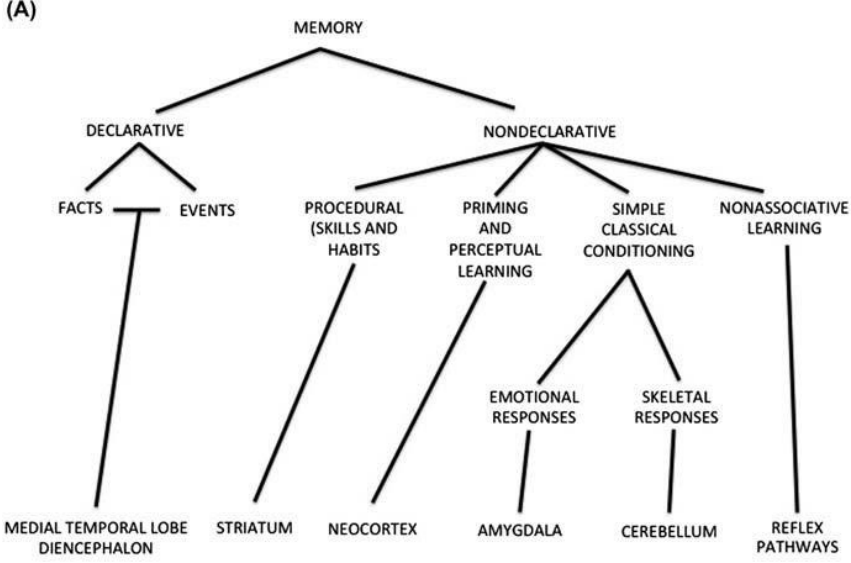
- Supervised learning
  - Learn from training data, labels.

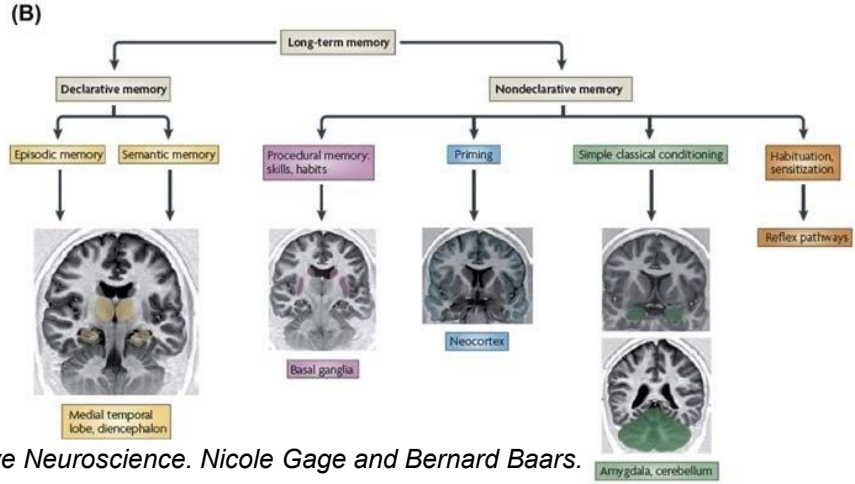
How do humans learn?

Process of transforming temporary synaptic connectivities into longer-lasting memories.

- **Episodic**: concious episodes linked to a place and time
- **Semantic**: general knowledge
- **Procedural**: skills and habits, often unconcious

Implicit (playing)
vs
Explicit (studying)





Fundamentals of Cognitive Neuroscience. Nicole Gage and Bernard Baars.

EPISODIC LEARNING AND MEMORY

Episodic memories may be learned simply by paying attention to some conscious event. Episodic learning can either happen *intentionally*—when we try to learn—or *incidentally*—without deliberately trying to memorize anything.

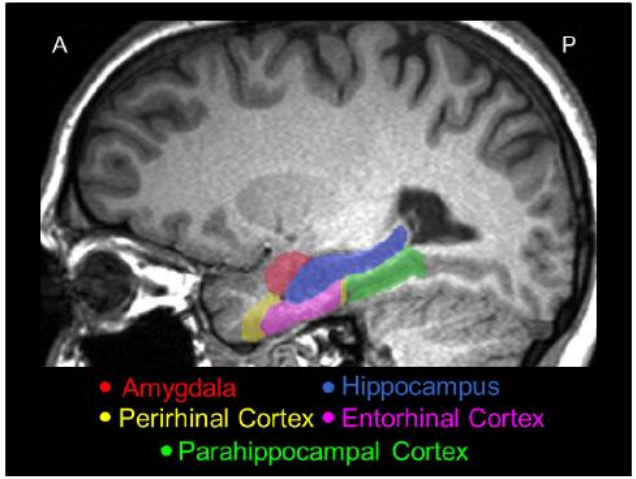
Human beings are extraordinarily accurate when we test for *recognition* of episodic memories, even when subjects have no deliberate intention to memorize it. That basic fact is easy to verify by checking a high school yearbook to see if you can recognize the photos of classmates you have not seen for a long time.

The high accuracy of episodic memory *as tested by recognition measures* suggests that our brains have a very large storage capacity for the things we pay attention to. Recognition measures are yes/no measures ("Yes, it is familiar" vs. "No, it is not"). It is harder to *recall* episodic memories from partial hints (what happened on your friend's birthday 5 years ago?). Recall accuracy is generally lower than recognition accuracy.

Episodic learning requires the hippocampus and its neighboring structures. A half-century of brain studies have shown that the hippocampal neighborhood (MTL) is needed for episodic learning—converting perceptual experiences into long-lasting episodic memories. The hippocampus is shaped like a thin tube inside each temporal lobe, but many small regions in this area play special roles. For that reason, the more inclusive term *MTL* is often used.

The basic question in episodic learning is simply this: How do sensory and other experiences turn into long-term autobiographical memories that may be needed years later?

MTL = medial temporal lobe



Fundamentals of Cognitive Neuroscience. Nicole Gage and Bernard Baars.

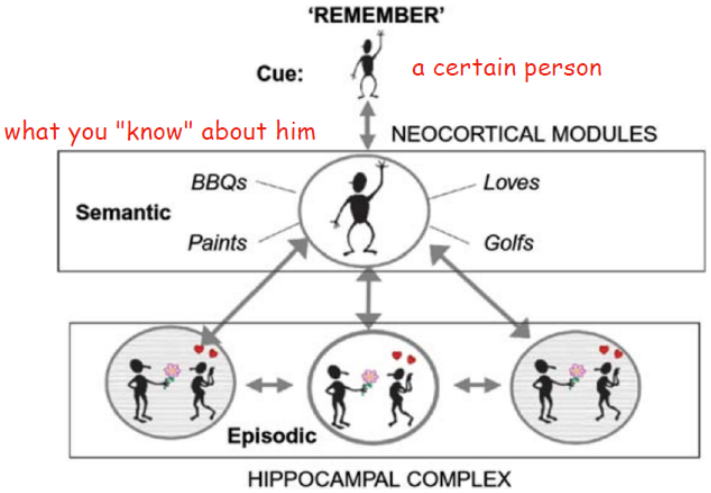
Semantic memory

Episodic memories typically:

1. have reference to oneself
2. are about a specific time
3. are *remembered* consciously, as if we reexperience them
4. can be forgotten
5. are *context-sensitive*, with remembered aspects of time, space, people, and circumstances

In contrast, semantic memories (Fig. 7.6) generally:

1. have reference to shared knowledge with others
2. are not organized around a specific time period
3. give a "feeling of knowing" rather than a fully conscious recollection of the original event
4. are less susceptible to forgetting than specific episodes
5. are relatively independent of context



(Fig. 7.6)

Learning disruptors

- Stress → strenghtens the amigdala, shifts priorities
- Sleep deprivation → school start times for adolescents is critical
- Noise
- Multitasking
- Interference**:** Biological learning is often disrupted by **proactive interference** (earlier info disrupts later learning) and **retroactive interference** (later info disrupts earlier memories).
- Functional Fixedness/aka tunnel vision: While expertise generally helps learning, it can lead to **mental rigidity**
- Fobias and PTSD

Fundamentals of Cognitive Neuroscience. Nicole Gage and Bernard Baars.

Meta-Gate

In another internal study of all Facebook users (see study 2.1), Meta researchers found that 3.1 percent of users met Meta's threshold for "severe problematic use." In internal correspondence responding to the study, Mark Zuckerberg himself stated that "3 percent of billions of people is a lot of people...it's millions of people." In fact, 3 percent of three billion people (the total number of monthly active Facebook users) is 90 million people who are struggling with severe problematic use.

Not as widely discussed, but of equal importance, is the finding that 55 percent of users exhibit at least mild problematic use, which corresponds to more than a billion people.

2017, Facebook's founding president, Sean Parker, stated, "The thought process that went into building these applications, Facebook being the first of them, ... was all about: 'How do we consume as much of your time and conscious attention as possible?'"

polarization, news consumption, well-being, and daily social interactions." Meta had run a pilot test of this "deactivation study" model with an unknown sample size–the results of these pilot tests led to a halting of the project. Meta partnered "with Nielsen and use[d] a combination of (a) surveys, (b) log data from Facebook and Instagram, and (c) usage data from smartphones." (p. 26) **Study uses log data.**

FINDINGS:

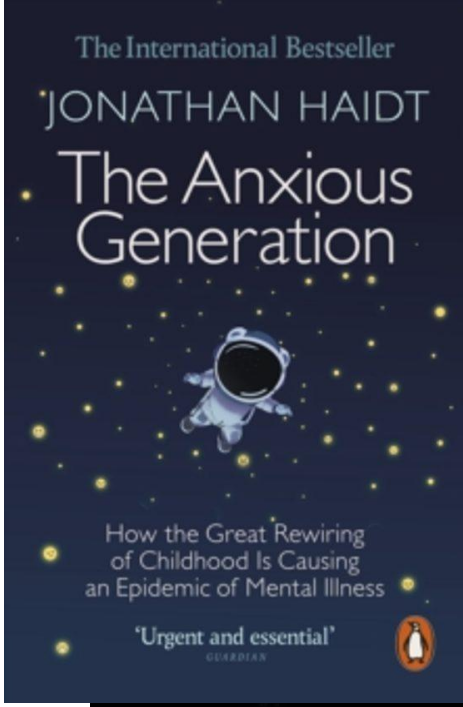
- The researchers found, in the pilot test, that "people who stopped using Facebook for a week reported lower feelings of depression, anxiety, loneliness, and social comparison." (p. 26)

https://metasinternalresearch.org/

Learning disruptors

- Short videos are causing a reward deficit disorder

-top 50% youngsters are more resistant to addiction/slot machines apps, they have the capacity to pay attention
-bottom 50% struggle



Reasoning as primary method for philosophical inquiry

- **Inductive**: the brain extracts statistical regularities from the environment to form concepts (and to hand with learning)

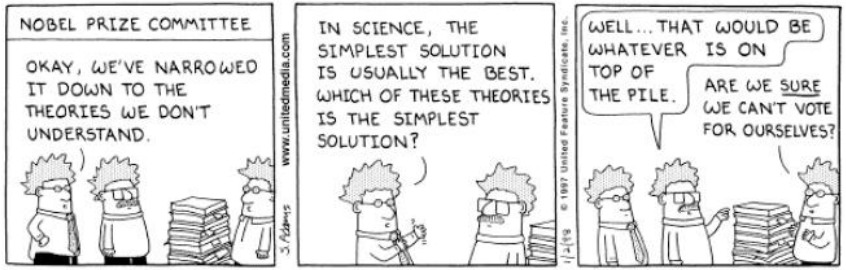
Training

- **Deductive reasoning** involves the application of the rules of logic to statements about the world: If this, then that.

Predicting

- **Abductive:** starts with a series of observations and finds the **simplest and most likely explanation.**

Interpretability



Caveat: ML is probabilistic, not cognitive

Inductive, deductive and abductive reasoning

- https://www.menti.com/alvvx8ovx543

Reasoning as primary method for philosophical inquiry

- **Deductive reasoning** involves the application of the rules of logic to statements about the world: If this, then that.

- **Inductive**: the brain extracts statistical regularities from the environment to form concepts.

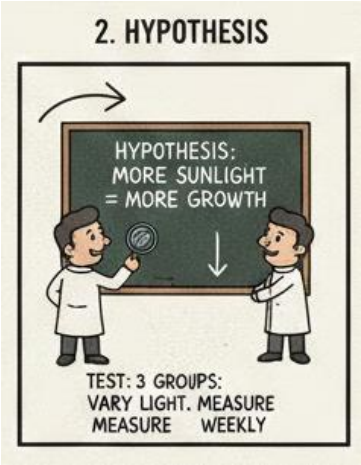
- **Abductive:** starts with a series of observations and finds the **simplest and most likely explanation.**

Q: problem?

- Critique: they do not seek empirical evidence → the scientific method

Reasoning: the scientific method

- We make observations that led us to ask a question
- Hypothesis: a testable explanation or reasonable prediction of what will happen
- Test the hypothesis, try to reject it.
- Report conclusions
- Use conclusions to generate new hypothesis



1. OBSERVATION & QUESTION

2. HYPOTHESIS & TEST

3. EXPERIMENT

3. EXPERIMENT

4. ANALYZE & CONCLUDE

Reasoning + scientific method

- Experimental philosophers and the Knobe effect.

Has the chairman harmed the environment intentionally?

82% yes

Has the chairman helped the environment intentionally?

23% yes

This conclusion, could not necessarily have been determined a priori using pure reasoning

Back to ML

- ML needs loads of training data
- Data augmentation -> apply small variations on existing training data
- Along with regularization techniques, avoid overfitting

The machine learning puzzle

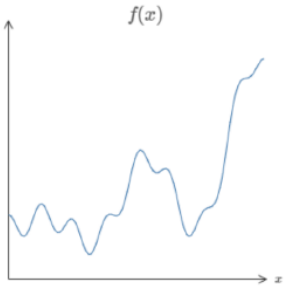
Three pieces:

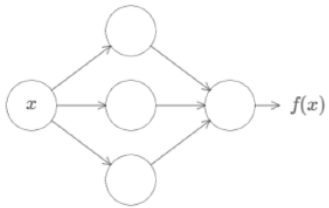
Approximation: does a NN that fits my training data?

Optimization: If it exists, can i find it?

Generalization: does it work well on unseen data?

- Neural networks can compute (almost) any function
- No matter the function, there is guaranteed to be a neural network that for every possible input x, the network closely approximates $f(x)$

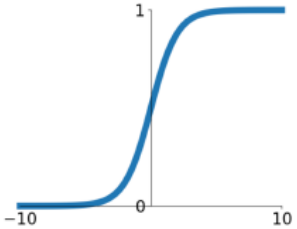




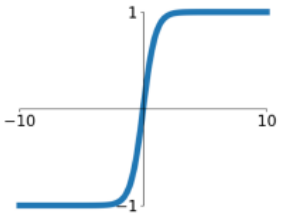
Activation functions

- Allow neurons to tackle non linear problems

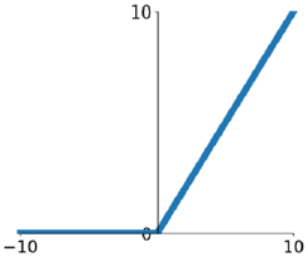
unit step function



Sigmoid



tanh(x)



ReLU
(Rectified Linear Unit)

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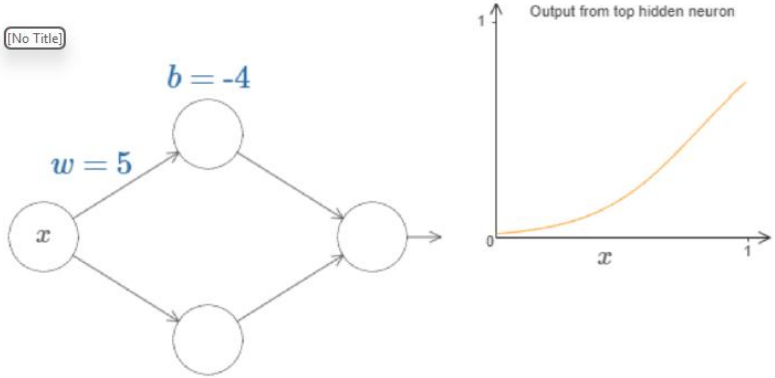
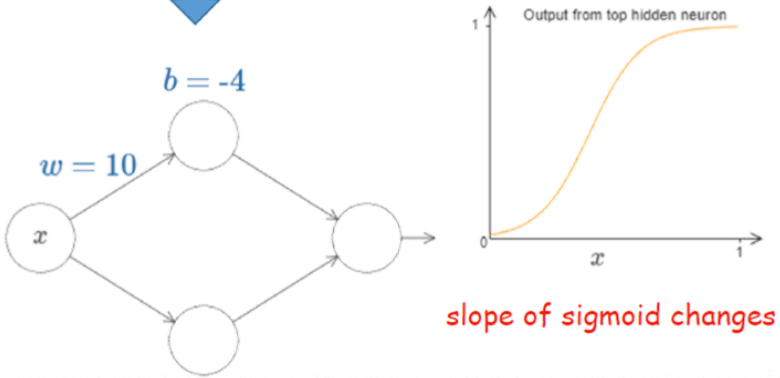
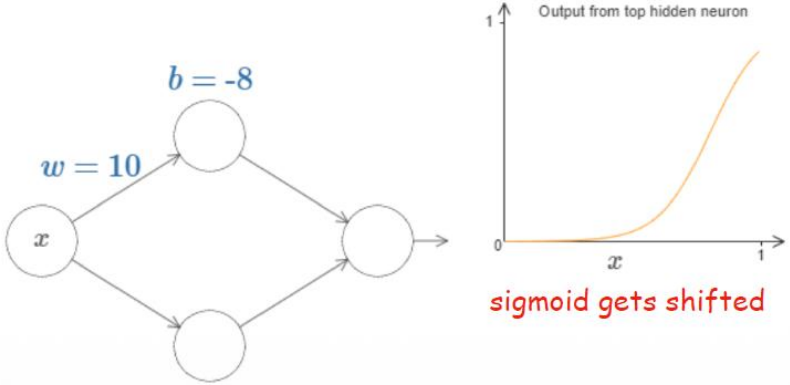


Increase *w*



slope of sigmoid changes

Decrease *b*



sigmoid gets shifted

Credit: G. Wolf (Yale), M. Nielsen

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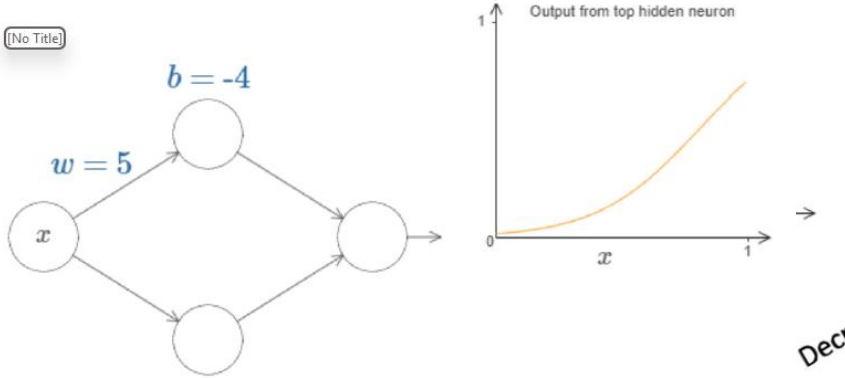
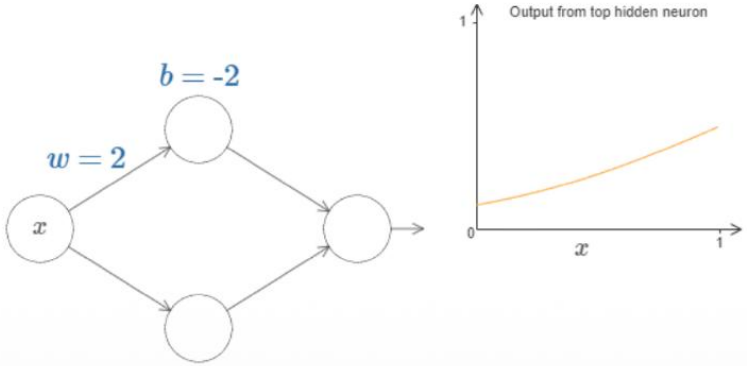
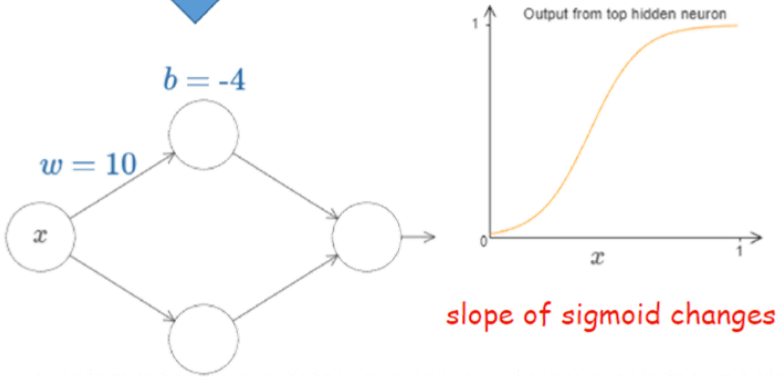
can make it very flat...



Decrease w

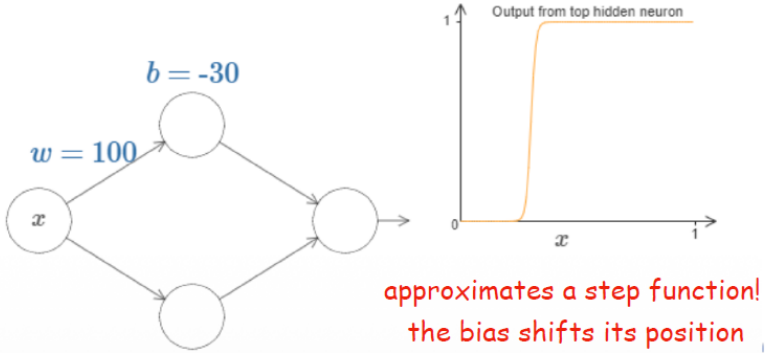


Increase w



slope of sigmoid changes

Increase w

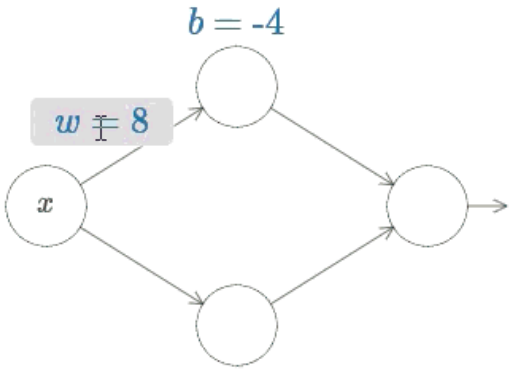


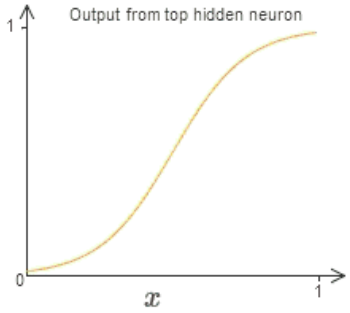
approximates a step function!
the bias shifts its position

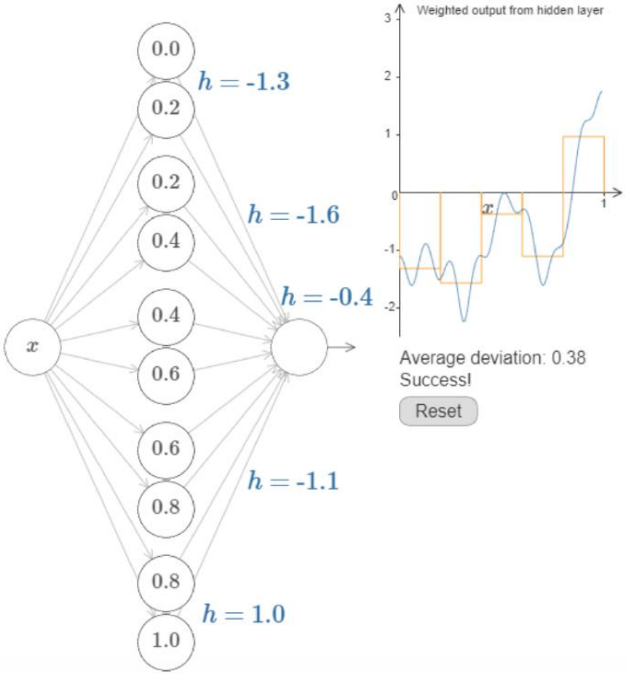
Credit: G. Wolf (Yale), M. Nielsen

Approximation

- We can simplify the activation function to step by setting large weights and modifying the bias

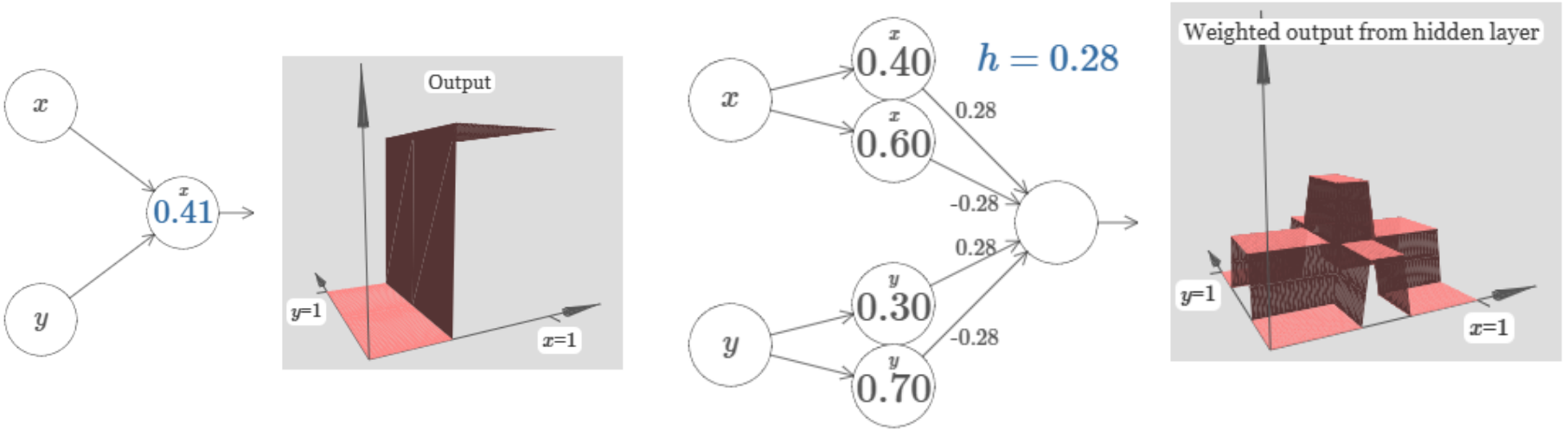






http://neuralnetworksanddeeplearning.com/chap4.html#universality_with_one_input_and_one_output

Approximation: two inputs



Approximation wrap up

- NN can represent any function
- Can the NN "learn" any function?
- How easy is it to learn a function? → NN are data hungry
- How well it performs in the real world? → Generalization

Generalization

Egyptian vulture ?

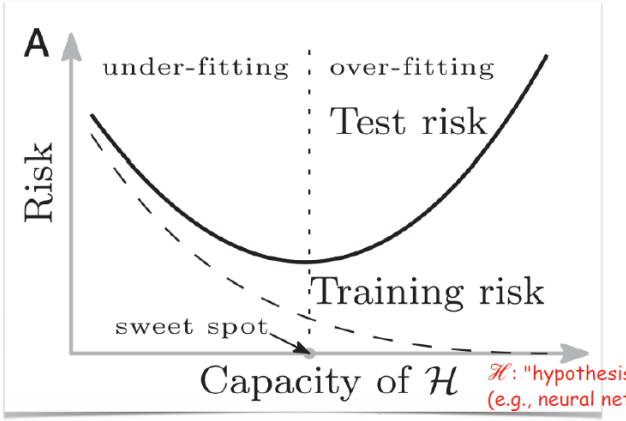
Egyptian vulture?

Overfitting and the bias-variance tradeoff

test error = train error + (test error - training error)

bias ***variance***

(rough approximation)



$\mathscr{H}$: "hypothesis space" that we search for predictors (e.g., neural nets)

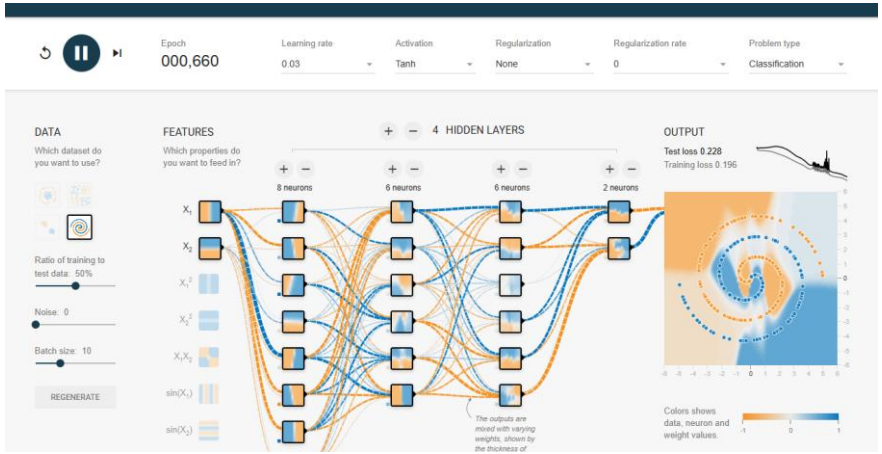
image: Belkin et al, 2019

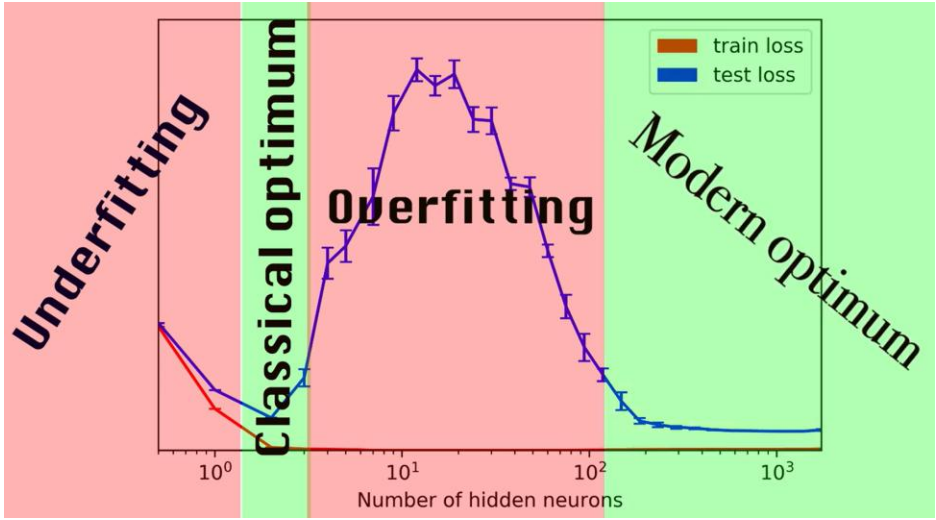
increasing the number of neurons/parameters →

Generalization

- Underfitting and overfitting

How can NN fit noisy data but generalize well?

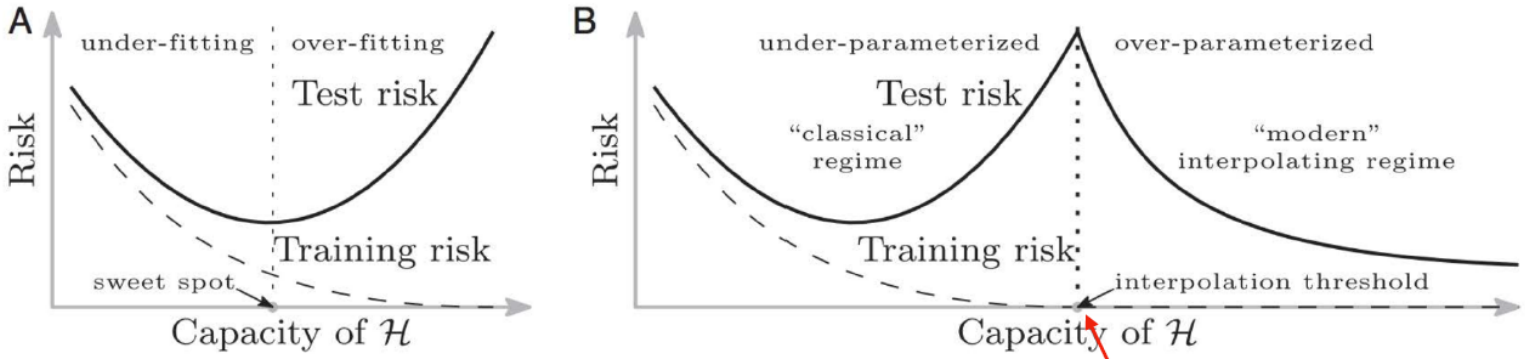




https://www.youtube.com/watch?v=Kih-VPHL3gA

Belkin: Reconciling modern ML with bias variance trade off: https://www.youtube.com/watch?v=ZAW9EyNo2fw

Genralization: double descent



Q: model can keep improving even after training error is minimized.... how?

weight regularization

Belkin, Hsu, Ma, Mandal, PNAS 2019

Caveats:
typically not feasible economically
needs special training conditions to actually happen in practice

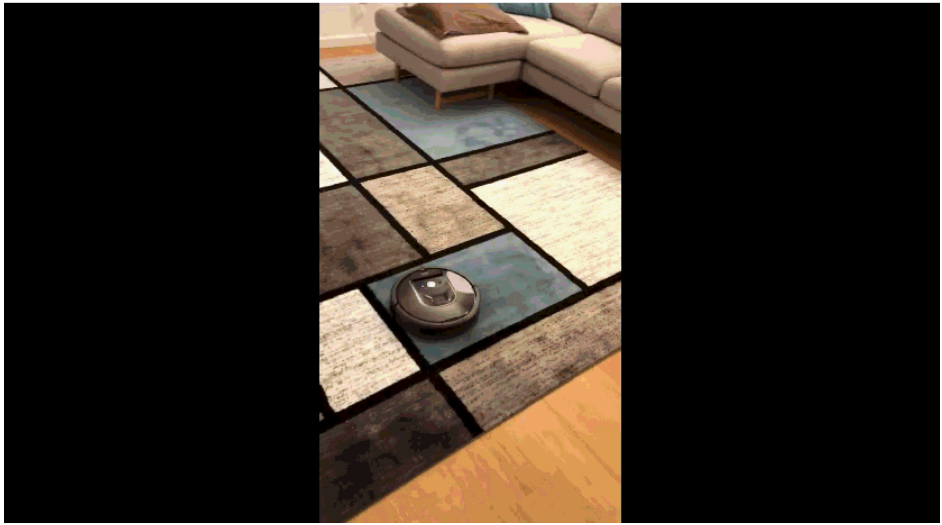
Generalization

The new Roomba uses AI to avoid smearing dog poop all over your house

Dmitry Krotov @DimaKrotov · Sep 10, 2021

Replying to @hardmaru

I wish they had also created a diverse dataset of rugs so that it didn't confuse black stripes with cliffs and I could finally get my entire house cleaned 😂



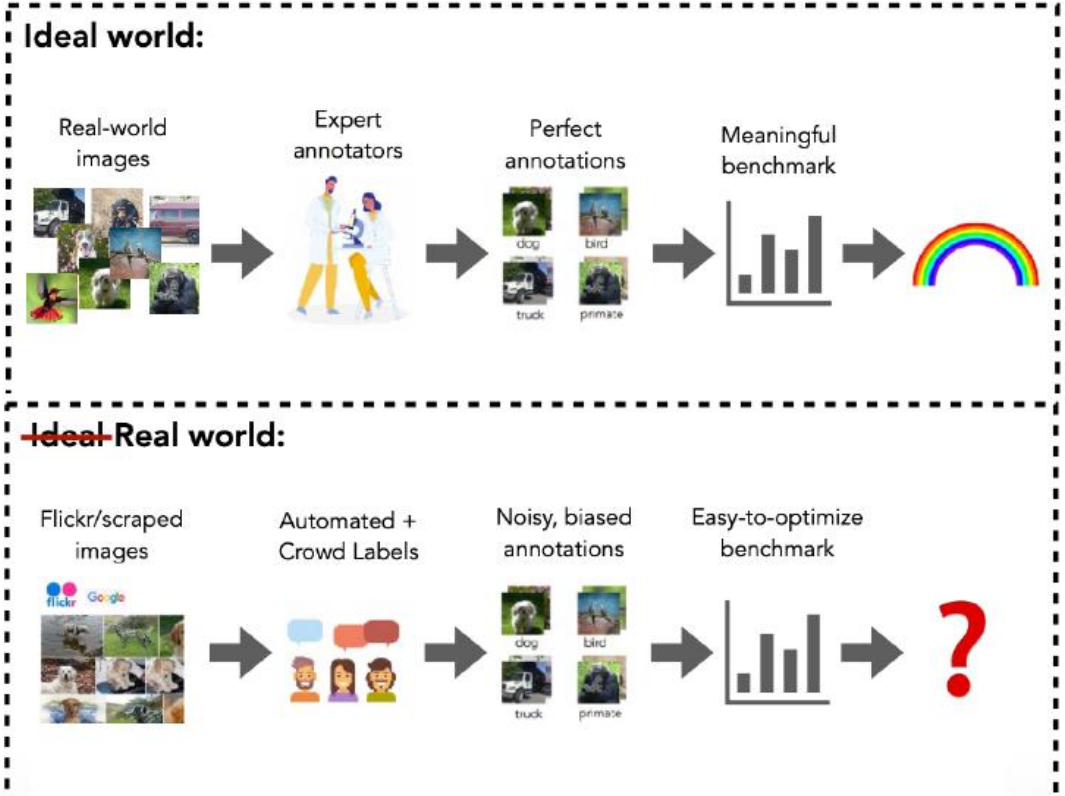
Bananas vs toasters: adversarial patches: https://www.youtube.com/watch?v=i1sp4X57TL4

Where do this correlations come from? Shortcuts

"Fish" from the ImageNet training set

[Esteva et al. 2017]

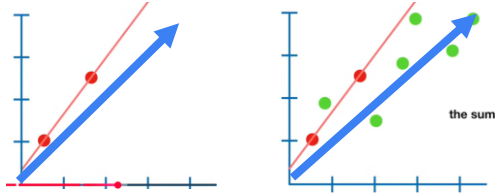
Higher chance of call a mole malignant if there was a rule in it



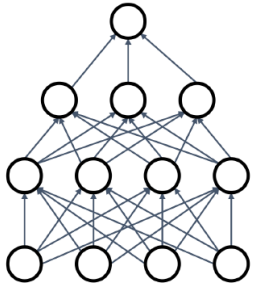
NN have trouble generalizing

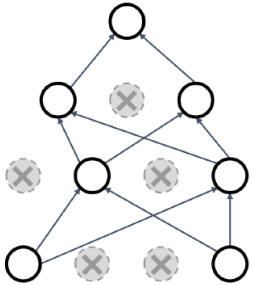
How can we help them? Regularization

- Add bias/penalty
- Drop-out

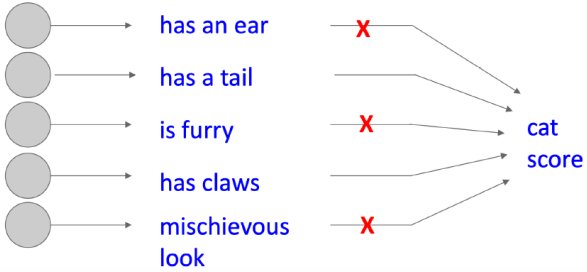


In each forward pass, randomly set some neurons to zero
Probability of dropping is a hyperparameter; 0.5 is common

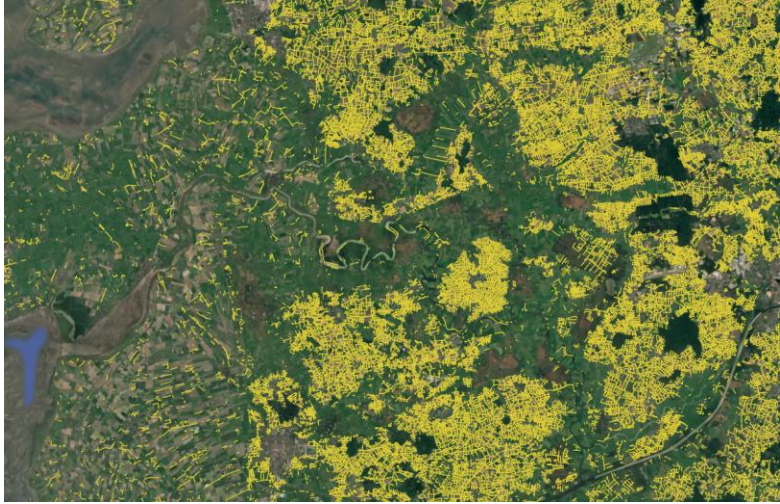




Forces the network to have a redundant representation; Prevents **co-adaptation** of features

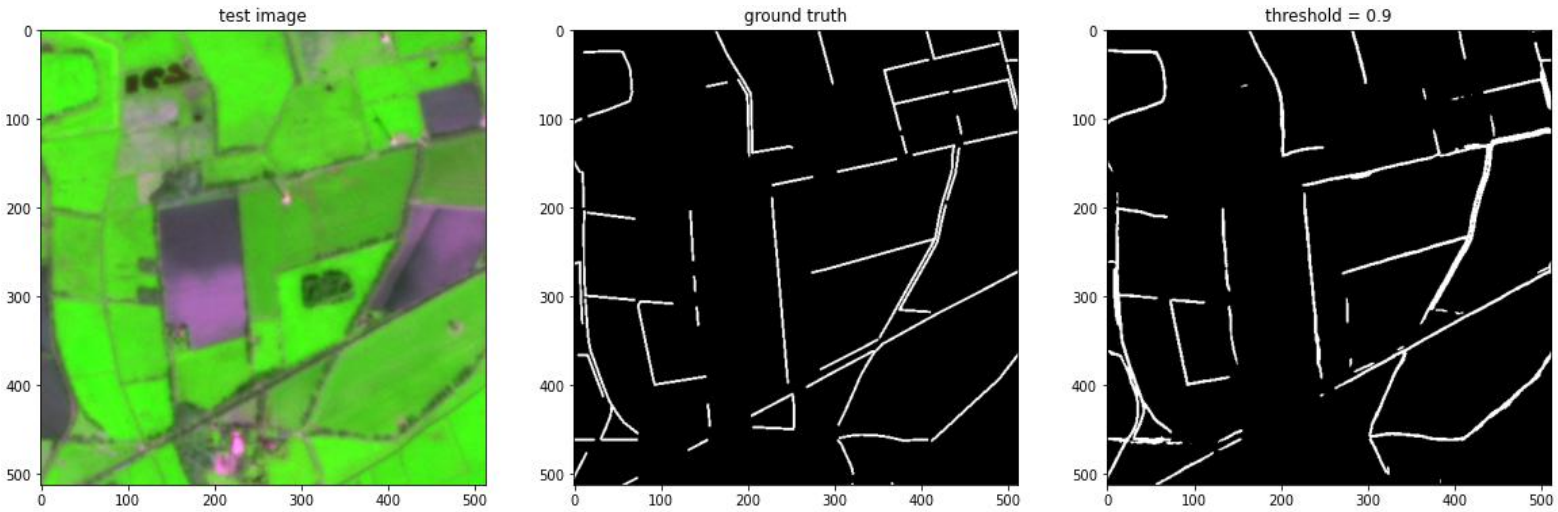


NN have trouble generalizing



How can we help them? Data augmentation I

- Flip, turn or tweak contrast or light improves generalization in image recognition



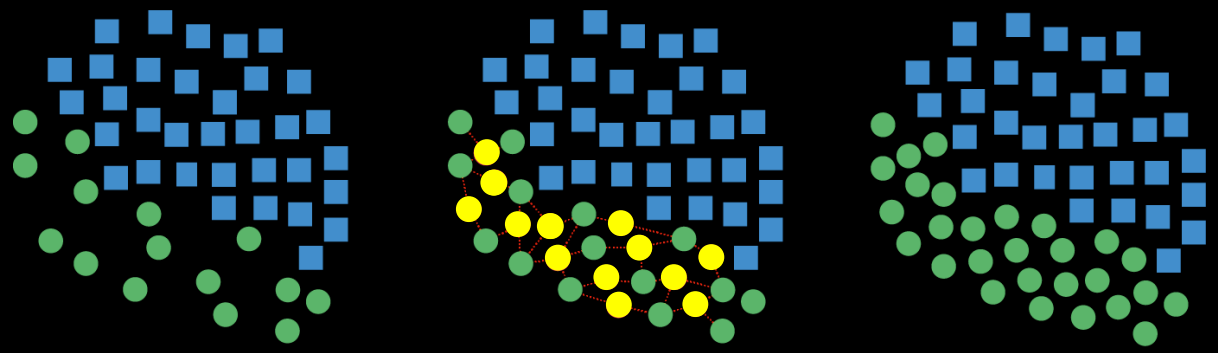
https://medium.com/@parthdholakiya180/smote-synthetic-minority-over-sampling-technique-4d5a5d69d720

Data augmentation II

Oversampling

Balance the dataset generating new rare samples by using:

SMOTE (Synthetic Minority Over-sampling TEchnique):



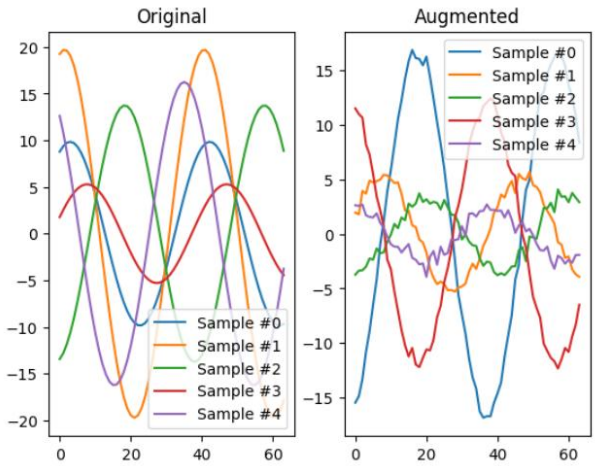
Original dataset

Generating Sample

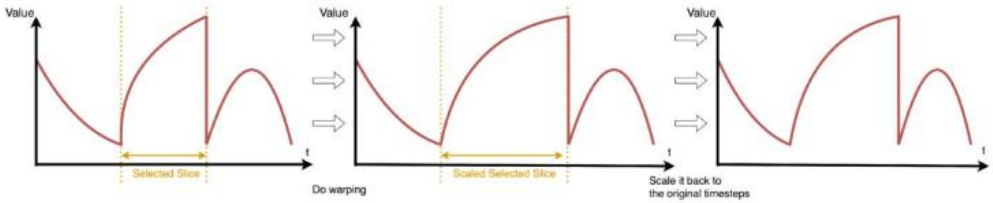
Resampled Dataset

Data augmentation III: time series

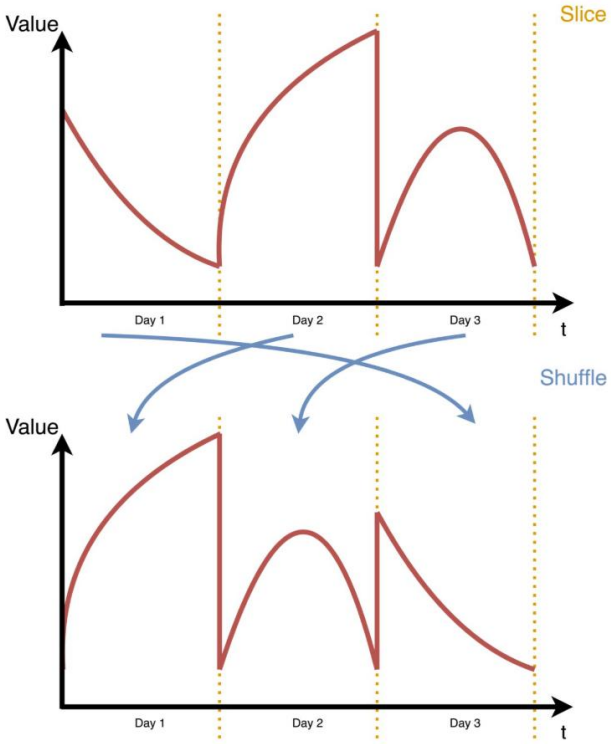
Introduce noise



Window warping



Split and shuffle



Source: Time Series Augmentations | Towards Data Science

How do humans generalize?

- Similarity based judgement: humans assess how closely a new situation resembles a **familiar** one in a psychological scape relying on propotypes

- **Rule** based application learnt through induction

- **Abstract structure mapping**: we identify the underlying abstract relational structure, rather than just its concrete features

How do humans generalize?

Two stage process:

1.- Determining **relevance**: the brain automatically determines what features are relevant.

2.- Refining **representations**: Through experience, mental representations of these features become more precise, allowing for more efficient, abstract and accurate future generalizations

Opposite of Deep learning: the better we understand something, the better we can abstract and generalize to novel scenarios

Cognitive components

Metageneralization: we can flexibly switch between different strategies such as **compositional** (breaking down a problem, useful in pseudocode) or **holistic** (conjunctive) approaches depending on the structure of the task.

Human-like systematic generalization through a meta-learning neural network

How do humans generalize?

Brenden M. Lake & Marco Baroni

Nature **623**, 115–121 (2023) | Cite this article

https://www.nature.com/articles/s41586-023-06668-3

Cognitive components

Efficient coding: we aim for mental efficiency by distilling information into simple, essential forms, often priorizing a small subset of features (color, shape of ripe fruit) to guide decisions

Humans learn generalizable representations through efficient coding

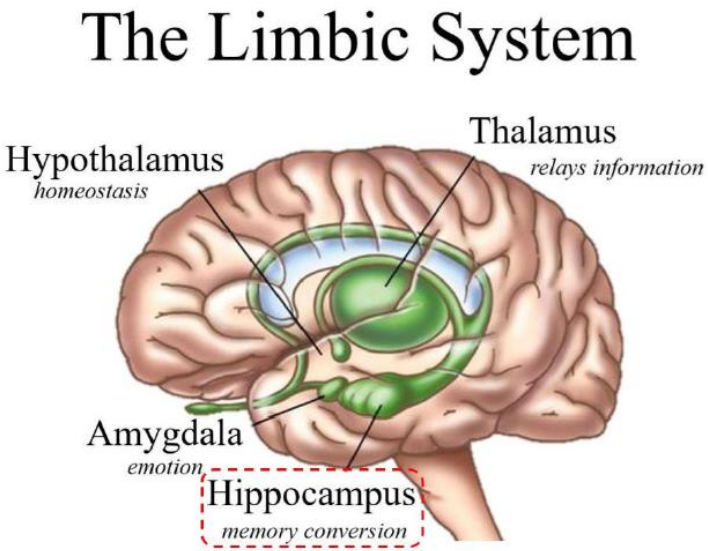
Zeming Fang & Chris R. Sims

Nature Communications **16**, Article number: 3989 (2025) | Cite this article

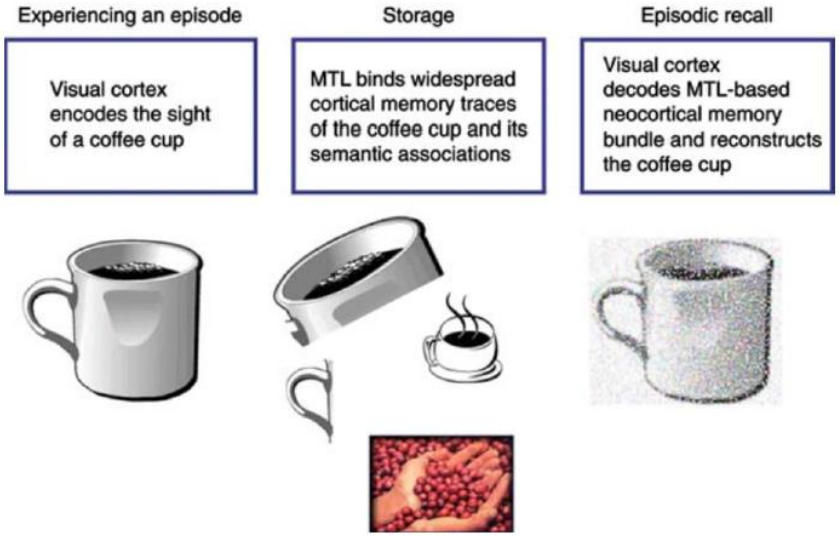
https://www.nature.com/articles/s41467-025-58848-6

Neural components

Hippocampus: rapid, episodic generalization is thought to rely here, which reactivates and integrates existing memories with new experiences



Recalling past experiences



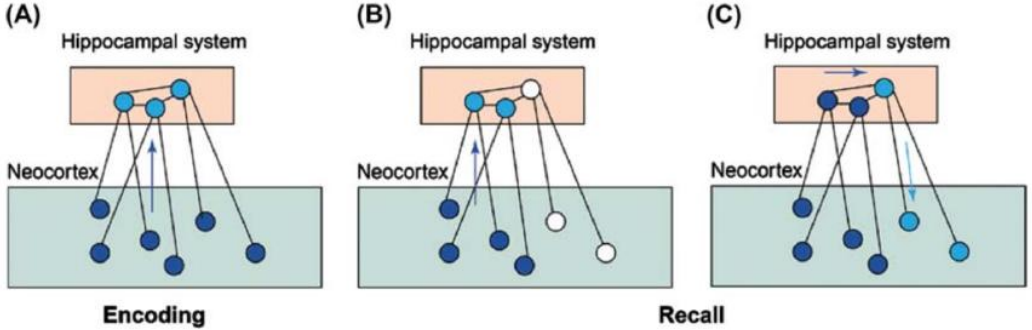


FIGURE 7.5 Hippocampus and cortex. The hippocampal area supports temporary episodic memories that activate and are stored in the cortex. In this cartoon, we show stages of encoding and recall and the roles of the hippocampal system and the cortex. (A) An episodic memory is formed (Encoded) in the cortex and gives rise to a pattern of activity with the hippocampal system. (B) During Recall, part of the pattern is active in the cortex, reactivating part or all of the hippocampal pattern. (C) The reactivation of the hippocampal pattern will, in turn, reactivate the remaining cortical pattern. *Source: Gluck et al. (2003).*

Fundamentals of Cognitive Neuroscience. Nicole Gage and Bernard Baars.